

Preferences and Reward Models

DATA 443/643 · Week 9, note 9.1

Week 9

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2025 L10 (Bradley–Terry RM; best-of-n is a baseline). Original slides; that course is not copied.

Lecture plan

- 1 A pair is an order, not a grade
- 2 Logistic loss on the difference
- 3 Chance is $\log 2$
- 4 Best-of-n uses the RM without PPO

Key goal: only Δ enters. r_ϕ is a fit to your raters, not “helpfulness.”

First time: what / why / formula

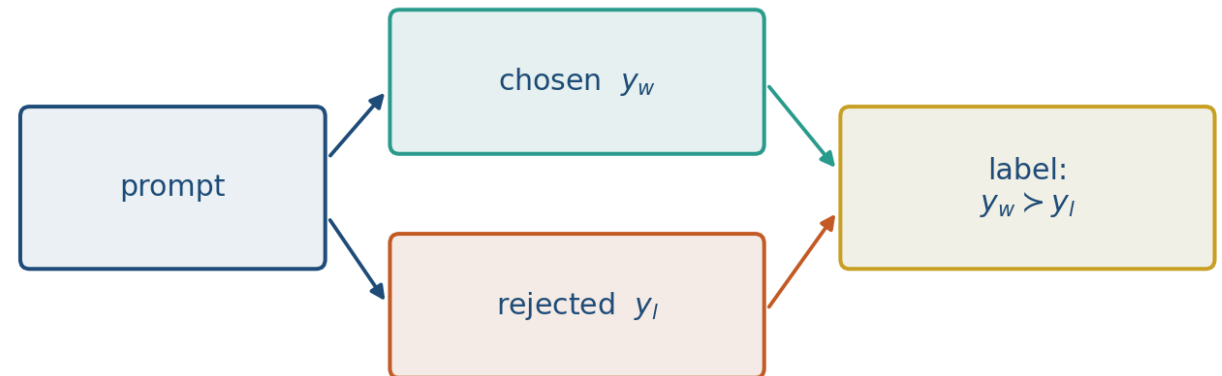
- What / why. Humans (or a teacher) pick y_w over y_l for prompt x . Bradley–Terry turns that into a logistic loss on $r(x, y)$. SFT copies a single answer. Preferences say “this is better than that” without writing a scalar label by hand.
- Architecture. Usually the base LM plus a scalar head, or a small classifier on the last token. Best-of- n : sample n , pick $\arg \max r$, no PPO yet.
- Formula. $p(y_w \succ y_l \mid x) = \sigma(r(x, y_w) - r(x, y_l))$. Loss $-\log \sigma(\Delta r)$.
- Tradeoffs. + Pairwise data is easier than absolute scores. – Reward hacking (next notes); RM error becomes policy error; best-of- n is compute at decode.

How. Collect pairs. Train r . Chance loss is $\log 2$. Lab 9 is a tiny head.

Chosen versus rejected

- Label: $y_w \succ y_l$ given prompt x .
- Both can be fluent. One is better for the rater.
- If both are wrong, the RM still learns a relative order.

One preference row is a comparison, not a score out of 100



Bradley–Terry on a pair

$$\mathcal{L} = -\log \sigma(r_w - r_l)$$

$\sigma(0) = 1 / 2$, so chance is $\log 2 \approx 0.693$.

Only $\Delta = r_w - r_l$ enters.
Absolute “7/10” does not.

Initialize the RM from SFT so it already speaks assistant.

Write the three losses

- $\Delta = 2$: $\sigma \approx 0.881$, loss ≈ 0.127 .
- $\Delta = -2$: the RM is wrong; loss ≈ 2.13 .
- Lab 9 should print both the easy pair and a flip.

Chance is $\log 2 \approx 0.693$. Only $\Delta = r_w - r_l$ enters.

Δ	$\sigma(\Delta)$	loss
0	0.500	$\log 2 \approx 0.693$
2	0.881	0.127
-2	0.119	2.13

The RM is already useful

Best-of-n

Sample n from SFT, return the RM's favorite.
AlpacaFarm: a strong baseline. Extra cost is n forwards, not a PPO loop.

PPO (next note)

Trains a new π_θ . Needs on-policy samples. Same RM.
Different job.

What to take from this lecture

Pairs

An order, not a score out of 100.

Held-out

How often $r_w > r_l$ on pairs the RM did not train on.

Not RLHF

Best-of-n uses the RM. It does not train a policy.